Vitor Gama

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Morgantown, WV, USA

Professional Summary

Innovative and detail-oriented Chemical Engineer with expertise in process modeling, membrane separation, and direct air capture (DAC). Adept at applying cutting-edge research and technical solutions to optimize efficiency, quality, and integrity in chemical processes. Passionate about leveraging ingenuity and accountability to solve complex engineering challenges while maintaining strong leadership and collaborative skills.

Key Skills

- **Process Optimization & Efficiency:** Developed process models and simulation frameworks to enhance membrane DAC system performance.
- **Technical Expertise & Ingenuity:** Extensive experience with Aspen Plus, Aspen Custom Modeler, HYSYS, and Python for process simulation and data analysis.
- **Accountability & Leadership:** Managed interdisciplinary projects, ensuring on-time and high-quality deliverables.
- **Safety & Compliance:** Adhered to best practices in chemical process safety and environmental regulations.
- **Software Development & Automation:** Designed cost estimation and control systems using C# and Python for chemical processes.

Education

West Virginia University — Morgantown, WV
Ph.D. student, Chemical Engineering — *Aug. 2021 – Dec. 2025 (Expected)*
GPA: 3.75/4.00

Federal University of Campina Grande — Campina Grande, Paraiba, Brazil
M.Sc., Chemical Engineering — *Sept. 2018 – April 2021*
GPA: 9.14/10.00

Federal University of Campina Grande — Campina Grande, Paraiba, Brazil
B.Sc., Chemical Engineering — *May 2013 – Aug. 2018*
GPA: 8.18/10.00

Research Experience

West Virginia University — Morgantown, WV
Graduate Research Assistant (Ph.D.) — *Aug. 2021 – Present*

- Developed simulation and operability analysis frameworks for gas separation membranes in Direct Air Capture (DAC) with Carbon Capture Utilization and Storage (CCUS).
- Applied inverse design and process operability analysis to optimize system performance, improving efficiency by **[XX]**%.
- Published in peer-reviewed journals, demonstrating commitment to quality and research integrity.

Federal University of Campina Grande — Campina Grande, Paraiba, Brazil
Graduate Research Assistant (M.Sc.) — *Sept. 2018 – April 2021*

- Designed and implemented **CostApp**, a cost estimation tool improving engineering decision-making efficiency.
- Developed process control strategies in collaboration with Petrobras, focusing on automation and efficiency enhancements.

Awards & Achievements

- AIChE Environmental Division Graduate Paper Award (2024) – 2nd place for research on membrane DAC process operability.
- Jack and Marietta Mullenger Fellowship (2024) – Recognized for academic excellence and research contributions.
- Statler College PhD Recruitment Fellowship (2021).

Technical Proficiencies

- **Programming:** Python, MATLAB, C#, JavaScript, LaTeX.
- **Process Simulation:** Aspen Plus, HYSYS, PRO/II, AVEVA Process Simulation, ChemCad.
- **Version Control:** Git, GitHub, GitLab.
- **Languages:** English, Portuguese.

Professional Leadership & Service

- President, Brazilian Student Association (2022 – 2024).
- Member, Statler College DEI Student Committee (2022 – 2023).
- Research Group Leader, CODES (2022 – Present).